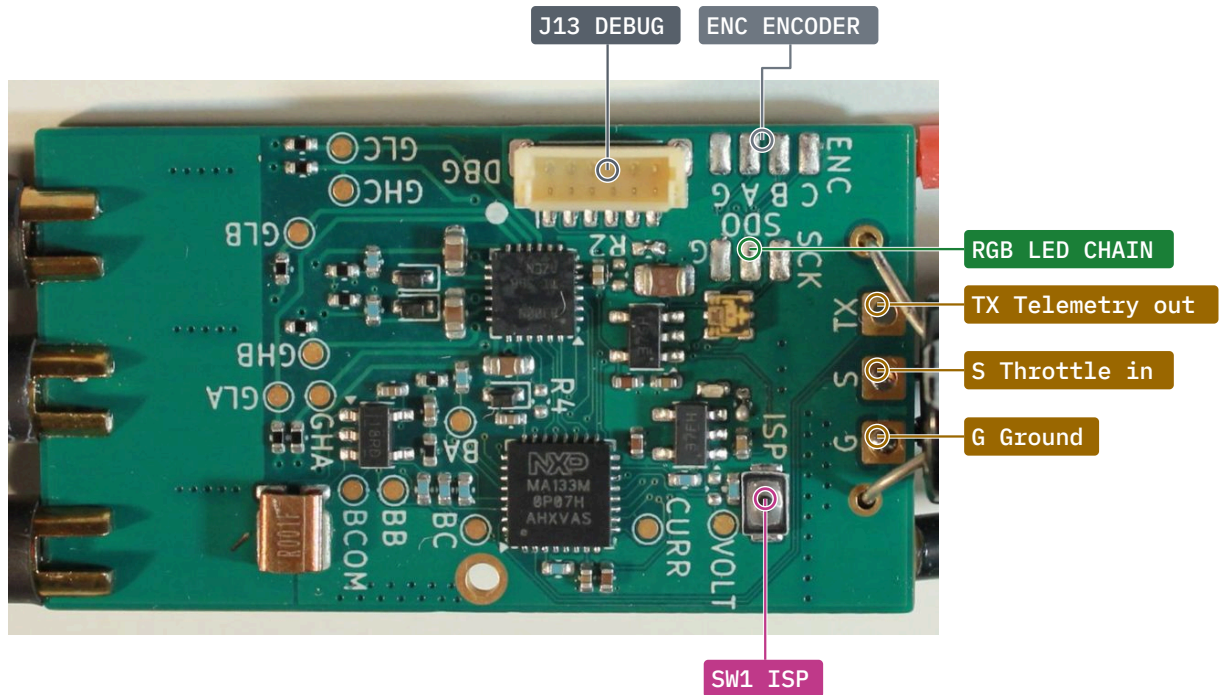


MR-ESC-MCXA-AM32 hardware reference

Connections of the NXP MR-ESC-MCXA-AM32, a small brushless motor controller (ESC) built around the MCXA153 or MCXA133 MCU and running the open AM32 firmware. All data comes from the KiCad design in the [MR-ESC-MCXA-AM32 repository](#).

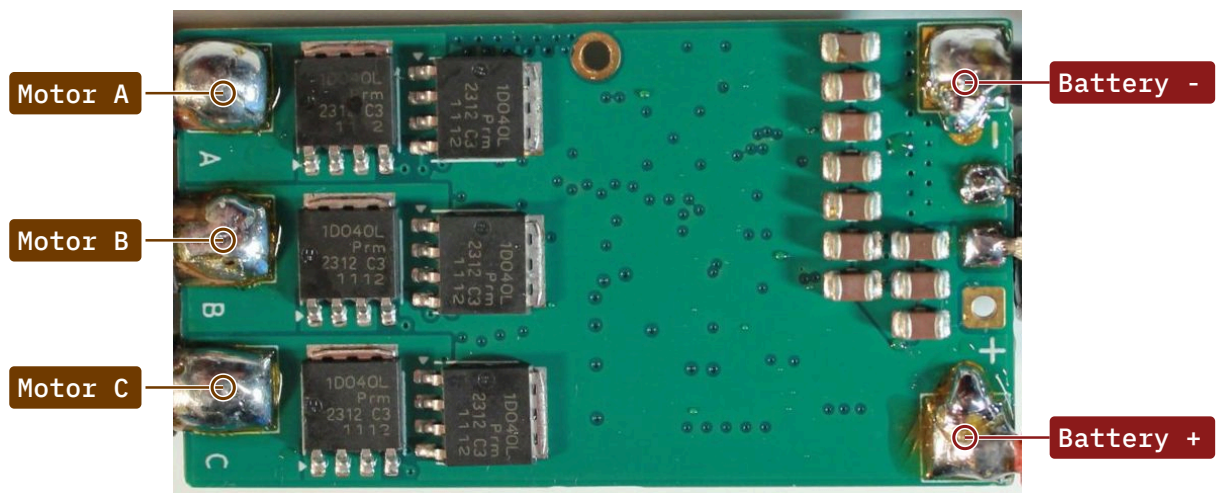
Each card shows, per pin, the **function**, the **net name in the schematic** and the **MCU pad** the signal ends up on. Most connections are solder pads; the only connector is the debug port.

Board overview



Not visible here: [PWR BATTERY](#) (bottom view) · [MOTOR PHASES](#) (bottom view)

Bottom side, with the battery and motor pads:



Connector index

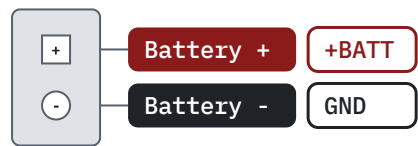
Ref	Function	Connector	Details
ESC ENC ENCODER	Spare MCU pins for a quadrature encoder, solder pads	Four pads marked A, B, I and G, 3.3 V logic	
ESC MOTOR PHASES	Three motor phase outputs	Three solder pads on the bottom side, marked A, B and C	
ESC PWR BATTERY	Battery input, 6 to 24 V (2S to 6S LiPo)	Two solder pads on the bottom side, marked + and -	
ESC RC SIGNAL	Throttle input and telemetry output, solder pads	Three wire pads marked S, G and TX, 3.3 V logic	
ESC RGB LED CHAIN	Output of the on-board RGB LED chain, solder pads	Three pads marked SDO, SCK and G, 3.3 V logic	
ESC J13 DEBUG	SWD debug and console (DS-009 debug mini)	JST-SH 1x6 (BM06B-SRSS-TB)	

Power and motor

ESC PWR BATTERY

PWR BATTERY

Two solder pads on the bottom side, marked + and -



Solder the battery leads here. Input range is 6 to 24 V, so 2S to 6S LiPo. The 5 V and 3.3 V rails come from two TPS7B92 linear regulators on this input; below about 6 V the 5 V rail drops out. All parts on the battery rail are rated 35 V or more (capacitors 35 V, regulators 40 V, FETs 80 V), which leaves margin for a full 6S pack at 25.2 V. Keep the 470 μ F capacitor across the pads when you use long battery leads.

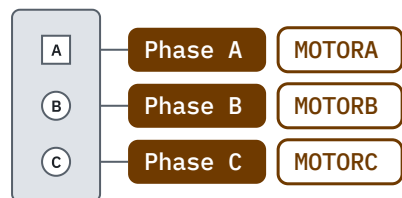
Caution

There is no reverse polarity protection. Check + and - before you connect a battery.

ESC MOTOR PHASES

MOTOR PHASES

Three solder pads on the bottom side, marked A, B and C



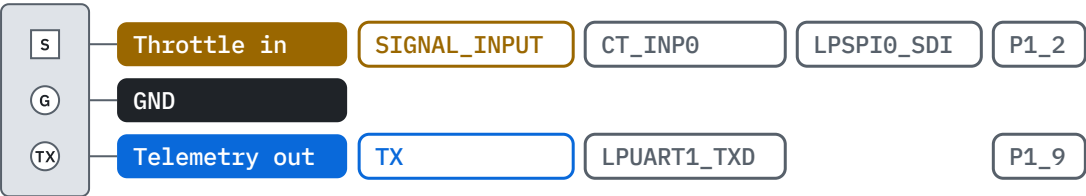
Each phase is a half bridge of two PSMN3R3-80 FETs (80 V), driven by the gate driver U2. Swap any two phases to reverse the motor direction. The phase voltages also feed the back-EMF comparators of the MCU, which is how the ESC senses the rotor position.

Control

ESC RC SIGNAL

RC SIGNAL

Three wire pads marked S, G and TX, 3.3 V logic



S is the throttle signal from the flight controller. The AM32 port for this MCU detects the input type by itself and supports DShot300, DShot600 (both also as bidirectional DShot) and normal servo PWM (1 to 2 ms, up to 400 Hz). DShot150 and other protocols are not supported. AM32 reads the signal with a CTIMER0 capture input and, for bidirectional DShot, switches the same pin to LPSPI0 to send the RPM telemetry back on the signal wire. TX is the AM32 serial telemetry output on LPUART1, 115200 baud, for a flight controller ESC telemetry input.

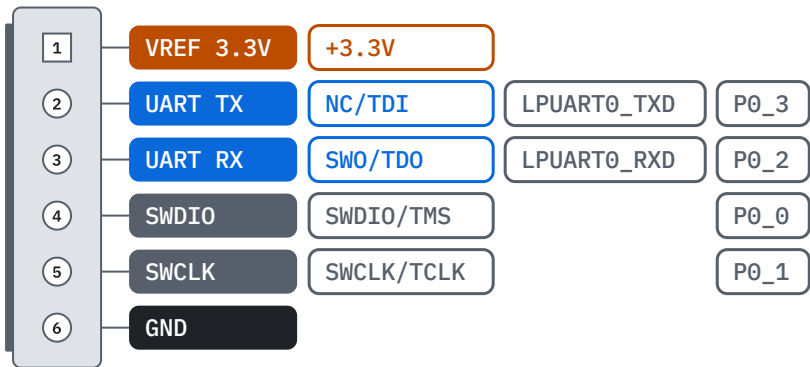
Note

The signal input is 3.3 V logic. Do not connect it to a 5 V PWM source without a level shifter.

ESC J13 DEBUG

J13 DEBUG

JST-SH 1x6 (BM06B-SRSS-TB) · DS-009 debug mini · 3.3 V logic



Flash and debug the MCU over SWD with any probe that supports the DS-009 debug mini cable. Pins 2 and 3 are LPUART0 on the MCU, free for a console; AM32 sends its telemetry on LPUART1 (the TX pad) instead. Pin 1 is a 3.3 V reference output for the probe.

Warning

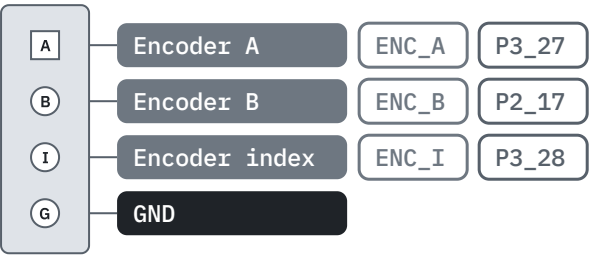
Pin 1 is a reference output, not a supply input. Never power the board from it.

Expansion pads

ESC ENC ENCODER

ENC ENCODER

Four pads marked A, B, I and G, 3.3 V logic



Prepared for an encoder on wheeled robots, where the ESC needs odometry. The AM32 firmware does not use them yet; it only drives A and I as debug outputs. Nothing else on the board is connected to these pins.

ESC RGB LED CHAIN

RGB LED CHAIN

Three pads marked SDO, SCK and G, 3.3 V logic



The MCU drives one APA102-type RGB LED (U5) on LPSP11. These pads are the data and clock outputs of that LED, so more LEDs of the same type can be chained from here.

Test pads

Thirteen round pads on the top side, named on the silkscreen. They are probe points for a scope or meter, not for wires.

Pad	Net	MCU pin	What you see there
GHA, GHB, GHC	GHA, GHB, GHC	P3_1, P3_9, P3_11 drive them	Gate of the high side FET of phase A, B, C. Output of the gate driver U2.
GLA, GLB, GLC	GLA, GLB, GLC	P3_0, P3_8, P3_10 drive them	Gate of the low side FET of phase A, B, C. Output of the gate driver U2.
BA, BB, BC	BEMFA, BEMFB, BEMFC	P1_0, P1_1, P1_3	Back-EMF of phase A, B, C, scaled down by a divider. The MCU comparator uses these to find the rotor position.
BCOM	BEMF_COMMON	P2_2, P2_3	Virtual star point: the three phase dividers joined. Reference for the back-EMF comparator.
VOLT	VOLTSENSE	P2_16 (ADC0 channel 6)	Battery voltage, scaled down by a divider.

Pad	Net	MCU pin	What you see there
CURR	CURRENT	P2_7 (ADC0 channel 7)	Output of the INA180 current amplifier: motor current as a voltage.
GND	GND		Ground for the probe.

Buttons and switches

Ref	Board	Function	Type	Notes
SW1 ISP	ESC	ISP boot button	Tactile switch	Hold it while power comes up to start the MCU boot ROM in ISP mode (P3_29 pulled low). Normal boot otherwise.

AM32 firmware

The board runs the open AM32 firmware. The firmware fork for this MCU, the hex files and how to flash them are in the [MR-ESC-MCXA-AM32 README](#).

Settings are changed with the [AM32 web configurator](#) over the signal wire, pad S. Connect it either through the flight controller (Betaflight passthrough) or with a serial link to a PC. No extra connector is needed. The input protocols the firmware accepts are listed on the [RC SIGNAL card](#).

Useful pages on the AM32 wiki:

- [ESC settings explained](#): what every setting does.
- [Recommended settings for freestyle drones](#).
- [Flashing a single ESC](#).
- [PC link with an Arduino](#): a serial link to the signal wire without a flight controller.
- [Hardware design notes](#): for changes to the board design.
- [EEPROM format](#): where the settings live in flash.

Tip

AM32 beeps through the motor at power up and after a settings change. Connect a motor before you look for the beep codes.

Notes

- Firmware, bootloader and flashing instructions are in the [MR-ESC-MCXA-AM32 README](#).
- The debug port follows the Dronecode [DS-009 connector standard](#) (debug mini), so a standard 6-pin debug cable fits.

Warning

The board has no reverse polarity protection and no input fuse. Wrong polarity on the battery pads destroys it.

Downloads

This reference as an [A4 PDF](#), and all cards on one A4 landscape sheet: [cheat sheet PDF](#).

NXP and the NXP logo are registered trademarks of NXP B.V. This document is maintained by the community and is not an official NXP publication.